# The Catalan generating function

Two derivations and their context-free grammars

This note gives complete proofs of Haiman's outer-block derivation and the standard first-return derivation, then expresses both as unambiguous context-free grammars. The Catalan calculations, including the square-root expansion, are self-contained. All power series are formal.

## 1. What is being counted?

A *parenthesis word* is a finite string of the symbols `(` and `)`. For a word $w$ of length $m$, define its height after $j$ symbols by

$$h_w(j) = \#\{\text{left parentheses among the first } j \text{ symbols}\} - \#\{\text{right parentheses among them}\}.$$

The word is *balanced* if $h_w(j) \geq 0$ for every $0 \leq j \leq m$ and $h_w(m) = 0$. Thus its length is $2n$ for some $n \geq 0$. Its *size* is $n$, the number of pairs, rather than the number of symbols. The empty word, denoted $\varepsilon$, is balanced and has size zero.

Let $C_n$ count balanced words of size $n$, and put $C(x) = \sum_{n\geq 0} C_n x^n$. In particular, $C_0 = 1$. For instance, the two words of size two are `(())` and `()()`.

**Theorem.** *The series $C(x)$ is the unique formal power series satisfying*

$$C(x) = 1 + xC(x)^2.$$

*Moreover,*

$$\boxed{C(x) = \frac{1 - \sqrt{1-4x}}{2x}} \qquad \textit{and} \qquad \boxed{C_n = \frac{1}{n+1}\binom{2n}{n} \quad (n \geq 0).}$$

*Here $\sqrt{1-4x}$ means the formal square root with constant term 1.*

## 2. The formal algebra we need

An element of $\mathbb{Q}[[x]]$ is a sequence of rational coefficients written as $F(x) = \sum_{n\geq 0} f_n x^n$. Equality means equality of every coefficient. Multiplication is defined by the finite sums

$$[x^n](F(x)G(x)) = \sum_{i=0}^{n} f_i g_{n-i}.$$

Combinatorially, the right side counts ordered pairs of objects of total size $n$, when $f_i$ and $g_j$ count the two kinds of objects of sizes $i$ and $j$. The notation $[x^n]F$ means the coefficient of $x^n$ in $F$.

If $A(0) = 0$, then $A(x)^k$ has no terms of degree below $k$. Consequently $\sum_{k\geq 0} A(x)^k$ is well-defined: at any fixed degree only finitely many summands contribute. The finite identity

$$(1 - A)(1 + A + \cdots + A^N) = 1 - A^{N+1}$$

therefore proves, coefficient by coefficient, that

$$\sum_{k\geq 0} A(x)^k = \frac{1}{1 - A(x)}. \tag{1}$$

More generally, any series with nonzero constant term is invertible: write it as $c(1 - A)$ with $c \neq 0$ and $A(0) = 0$, and use (1).

## 3. The unique decomposition into outer blocks

Call a nonempty balanced word a *block* if its height is strictly positive except at its two endpoints. Equivalently, it has no return to height zero before its last symbol.

**Lemma 1.** *A word is a block if and only if it has the form (u) with u balanced. Every balanced word is a unique ordered concatenation of blocks. The empty word corresponds to the empty concatenation.*

*Proof.* If $u$ is balanced, enclosing it in one pair of parentheses raises all heights along $u$ by one. The resulting word has positive height until its last symbol, so it is a block.

Conversely, a block must start with `(` and end with `)`. After removing these symbols, every remaining prefix has height equal to an interior height of the block minus one, which is nonnegative. The remaining word ends at height zero and is therefore balanced.

Now take any nonempty balanced word $w$ of length $2n$. List all its return times to height zero:

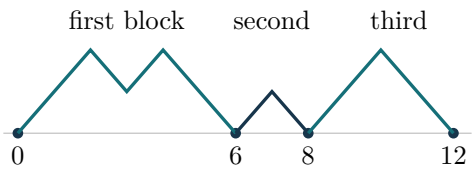
$$0 = t_0 < t_1 < \cdots < t_k = 2n.$$

Cut $w$ at these positions. Between consecutive return times its height is positive, so each resulting piece is a block. In any concatenation of blocks, the block endpoints are exactly the return times to zero: there are no such returns inside a block. Thus every cut is forced, which proves uniqueness. □

For example, the word `(()())()(())` splits as

$$\underbrace{(()())}_{\text{size 3}}\ \underbrace{()}_{\text{size 1}}\ \underbrace{(())}_{\text{size 2}}.$$

The picture records the height after each symbol. The dots on the baseline are exactly the permitted cut positions.




Let $A(x)$ count blocks by size. Enclosing a balanced word in a pair adds one to its size, and Lemma 1 gives a bijection, so

$$A(x) = \sum_{r \geq 1} C_{r-1} x^r = xC(x). \tag{2}$$

For fixed $k$, the product $A(x)^k$ counts ordered lists of $k$ blocks: its coefficient of $x^n$ is

$$\sum_{\substack{r_1+\cdots+r_k=n \\ r_1,\ldots,r_k \geq 1}} C_{r_1-1} \cdots C_{r_k-1}.$$

Uniqueness of the decomposition ensures that each balanced word is counted exactly once. The case $k = 0$ contributes the empty word. Since every block has size at least one, only $k \leq n$ can contribute at degree $n$. Hence (1) and (2) give

$$C(x) = \sum_{k \geq 0} (xC(x))^k = \frac{1}{1 - xC(x)}. \tag{3}$$

The denominator has constant term 1, so multiplication by it is valid. We obtain $C - xC^2 = 1$, or equivalently $C = 1 + xC^2$.

## 4. Solving the equation inside formal power series

First, the equation determines the coefficients uniquely. Its constant term is $C_0 = 1$, and comparison at degree $n+1$ gives

$$C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i} \qquad (n \geq 0). \tag{4}$$

Thus each next coefficient is determined by earlier ones. Existence has already been established by the counting series above.

To solve explicitly, complete the square:

$$(1 - 2xC(x))^2 = 1 - 4xC(x) + 4x^2C(x)^2 = 1 - 4x,$$

where the last equality uses $C - xC^2 = 1$. Put $S(x) = 1 - 2xC(x)$. Then $S(0) = 1$ and $S(x)^2 = 1 - 4x$.

Such a square root is unique. Indeed, if $T(0) = 1$ and $T^2 = S^2$, then $(T - S)(T + S) = 0$. The series $T + S$ has nonzero constant term 2, so it is invertible; multiplying by its inverse gives $T = S$. It follows that

$$C(x) = \frac{1 - S(x)}{2x} = \frac{1 - \sqrt{1 - 4x}}{2x}. \tag{5}$$

This division by $x$ is legitimate because $1 - S(x)$ has constant term zero: dividing a series $\sum_{j \geq 1} b_j x^j$ by $x$ simply shifts all exponents down by one. In contrast, $1 + S(x)$ has constant term 2, so $(1 + S(x))/(2x)$ is not a power series. This explains the minus sign.

## 5. A proof of the square-root expansion

For a rational number $a$, define

$$\binom{a}{0} = 1, \qquad \binom{a}{m} = \frac{a(a-1)\cdots(a-m+1)}{m!} \quad (m \geq 1), \qquad B_a(z) = \sum_{m \geq 0} \binom{a}{m} z^m.$$

We will prove the specific formal binomial identity that is needed, rather than assume an infinite binomial theorem.

**Lemma 2.** *For rational $a, b$, one has $B_a(z)B_b(z) = B_{a+b}(z)$ in $\mathbb{Q}[[z]]$.*

*Proof.* Fix $m \geq 0$. For nonnegative integers $a, b$, choosing $m$ elements from the disjoint union of an $a$-element set and a $b$-element set gives

$$\sum_{i=0}^{m} \binom{a}{i} \binom{b}{m-i} = \binom{a+b}{m}. \tag{6}$$

Both sides are polynomials in $a, b$ over $\mathbb{Q}$, using the definition above. Their difference vanishes identically: for each nonnegative integer $b$, it is a polynomial in $a$ with infinitely many roots; then each of its coefficients, viewed as a polynomial in $b$, also has infinitely many roots. Therefore (6) holds for all rational $a, b$. It says exactly that the coefficients of $z^m$ in the claimed series identity agree. Since $m$ was arbitrary, the lemma follows. □

Taking $a = b = 1/2$ gives $B_{1/2}(z)^2 = B_1(z) = 1 + z$. Substituting $z = -4x$ is coefficientwise valid, and the resulting series has constant term 1. By the uniqueness just proved,

$$\sqrt{1 - 4x} = \sum_{m \geq 0} \binom{1/2}{m} (-4)^m x^m. \tag{7}$$

## 6. Extracting the Catalan numbers

Substitute (7) into (5). The constant terms cancel. Because division by $x$ shifts exponents down by one, the coefficient of $x^n$ comes from the term indexed by $n+1$:

$$C_n = -\frac{1}{2}(-4)^{n+1}\binom{1/2}{n+1}. \tag{8}$$

For $n \geq 0$, expand the binomial coefficient:

$$\binom{1/2}{n+1} = \frac{(1/2)\prod_{j=1}^{n}(1/2-j)}{(n+1)!} = \frac{(-1)^n\prod_{j=1}^{n}(2j-1)}{2^{n+1}(n+1)!}.$$

An empty product is 1, so these identities include $n = 0$. Inserting this into (8), the sign is $-(-1)^{n+1}(-1)^n = 1$ and the power of two is $2^{2n+2}/(2\cdot 2^{n+1}) = 2^n$. Thus

$$C_n = \frac{2^n\prod_{j=1}^{n}(2j-1)}{(n+1)!}.$$

Since the even factors of $(2n)!$ have product $2^n n!$, we have

$$(2n)! = 2^n n!\prod_{j=1}^{n}(2j-1).$$

Consequently,

$$\boxed{C_n = \frac{(2n)!}{n!(n+1)!} = \frac{1}{n+1}\binom{2n}{n}.}$$

This also gives $C_0 = 1$ and completes the proof of the theorem.

## 7. The standard derivation: first return and recurrence

We can derive the functional equation directly, without using the sequence principle. Every nonempty balanced word has a unique expression

$$(\,u\,)\,v,$$

where the displayed closing parenthesis is the first return to height zero. The word $u$ is balanced by the block lemma, and the suffix $v$ is balanced because it starts at height zero and inherits the prefix inequalities. Conversely, any ordered pair of balanced words $u, v$ gives a balanced word in this way, with that first return.

If the whole word has size $n+1$ and $u$ has size $i$, then $v$ has size $n-i$. There are $C_iC_{n-i}$ choices, and summing over $0 \leq i \leq n$ gives $C_{n+1} = \sum_{i=0}^{n} C_iC_{n-i}$, with $C_0 = 1$. Multiply by $x^{n+1}$ and sum over $n \geq 0$:

$$\begin{aligned} C(x) - 1 &= \sum_{n\geq 0}\sum_{i=0}^{n} C_iC_{n-i}x^{n+1} \\ &= x\sum_{i,j\geq 0} C_iC_jx^{i+j} = xC(x)^2. \end{aligned}$$

The reindexing $j = n-i$ is valid because each coefficient receives only finitely many terms. Hence $C = 1 + xC^2$, and Sections 4–6 yield

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}, \qquad C_n = \frac{1}{n+1}\binom{2n}{n}.$$

The recurrence begins

$$C_0, C_1, C_2, C_3, C_4, C_5, C_6 = 1, 1, 2, 5, 14, 42, 132.$$

The factor $x$ records the outer pair; the two factors $C(x)$ record the independently chosen words inside that pair and after it.

## 8. Balanced words as a context-free language

A *context-free grammar* consists of a finite set of terminal symbols, a finite set of nonterminal symbols, a start nonterminal, and production rules $A \to \alpha$. Here $A$ is one nonterminal and $\alpha$ is a finite string of terminals and nonterminals, possibly empty. A rule replaces an occurrence of $A$ by $\alpha$, regardless of its surrounding symbols. The generated *language* is the set of terminal words obtainable from the start symbol by finitely many replacements.

For balanced words, take terminals `(`, `)`, one nonterminal $S$, and the rules

$$\boxed{S \longrightarrow \varepsilon \;\mid\; (\,S\,)\,S.} \tag{9}$$

The vertical bar separates alternative rules, and $\varepsilon$ is the empty word. This language is called the *Dyck language with one type of parenthesis*.

**Why the grammar generates exactly the balanced words.** Every completed derivation gives a balanced word: the empty word is balanced, and balanced words $u, v$ give the balanced word $(u)v$. This proves the assertion by induction on the number of rule applications. Conversely, the first-return decomposition of any nonempty balanced word gives $(u)v$, where both $u$ and $v$ have smaller size. Induction on size supplies their derivations, and the second rule combines them. The empty word uses the first rule.

A *parse tree* records which rule is used at each nonterminal and the ordered subtrees for the symbols on its right side. A grammar is *unambiguous* if each generated word has exactly one parse tree. Equivalently, each word has one leftmost derivation, in which the leftmost nonterminal is always replaced. Arbitrary orders of rule application may differ even when the parse tree is the same.

**Why this grammar is unambiguous.** The empty word must use $S \to \varepsilon$. For a nonempty word, the root rule must be the second rule of (9). Its displayed closing parenthesis is forced to be the first return to height zero, so the two subwords $u, v$ are uniquely determined. Induction on size gives a unique parse for each. Thus the whole parse tree is unique.

The counting translation is now justified by an actual bijection:

| Grammar construction | Contribution by number of pairs |
|---|---|
| $S \to \varepsilon$ | 1 (one word of size zero) |
| $S \to (S)S$ | $xC(x)^2$ (one pair and two ordered choices) |
| The two alternatives | $C(x) = 1 + xC(x)^2$ |

If instead $z$ counts individual terminal symbols, put $D(z) = \sum_{w \text{ balanced}} z^{|w|}$. Then

$$D(z) = 1 + z^2 D(z)^2 = C(z^2).$$

The exponent 2 accounts for the two displayed parentheses.

**The sequence derivation as a grammar.** Introduce a nonterminal $B$ for one outer block and write

$$S \longrightarrow \varepsilon \mid BS, \qquad B \longrightarrow (S).$$

Lemma 1 proves unambiguity. The equations are $C = 1 + AC$ and $A = xC$, so $C = 1/(1 - A) = 1/(1 - xC)$. Repeatedly using $S \to BS$ lists the blocks; inlining the rule for $B$ gives exactly (9).

## 9. Why unambiguity is essential

The sum and product translation naturally counts *parse trees*. It counts distinct terminal words only when each word has one parse tree. A small example makes the distinction explicit. Consider

$$T \longrightarrow a \mid TT.$$

Its words are $a, a^2, a^3, \ldots$, exactly one of each positive length, so their generating function is $z/(1-z)$. But the rule translation gives

$$P(z) = z + P(z)^2, \qquad P(0) = 0.$$

For $a^3$, the root can split the word as $a \cdot a^2$ or as $a^2 \cdot a$, giving two parse trees. More generally, each parse tree is an ordered full binary tree, with terminal $a$ at each leaf. The equation above, or substitution of $P = zC(z)$, gives

$$P(z) = zC(z) = z + z^2 + 2z^3 + 5z^4 + 14z^5 + \cdots .$$

Thus $a^m$ has $C_{m-1}$ parse trees. The same Catalan numbers count different binary parenthesizations of a fixed ordered string.

There is a further problem with the familiar grammar $S \to \varepsilon \mid (S) \mid SS$ for balanced words: it has infinitely many parses of $\varepsilon$, obtained by inserting arbitrarily many $S \to SS$ steps before erasing every $S$. Its naive equation $F = 1 + xF + F^2$ therefore does not define a series of finite parse counts. The first-return grammar avoids both overcounting and such zero-size cycles.

## 10. The general algebraic connection

For a grammar with nonterminals $A_1, \ldots, A_r$, let $F_i(z)$ count parse trees rooted at $A_i$ by the length of their terminal word. Assume there are only finitely many such trees of each length. A rule whose right side contains $k$ terminals and nonterminal occurrences $A_{j_1}, \ldots, A_{j_t}$ contributes

$$z^k F_{j_1}(z) \cdots F_{j_t}(z).$$

Indeed, the rule supplies the $k$ terminal positions, and its ordered subtrees can be chosen independently. Summing over the rules with left side $A_i$ gives a finite system of polynomial equations for the $F_i$. For a reduced unambiguous grammar, these are also word-counting equations. Here reduced means that every nonterminal is reachable from the start and can produce a terminal word.

The broader statement is the **Chomsky–Schützenberger enumeration theorem**: if $L$ is generated by an unambiguous context-free grammar, then

$$F_L(z) = \sum_{w \in L} z^{|w|} \quad \text{is algebraic over } \mathbb{Q}(z).$$

That is, a nonzero polynomial $Q(z, y) \in \mathbb{Q}[z, y]$ satisfies $Q(z, F_L(z)) = 0$. This general theorem is stated here for context; its general proof is not needed for the complete Catalan proof above. In the present example the polynomial is explicit:

$$z^2 D(z)^2 - D(z) + 1 = 0 \quad \text{or, counting pairs,} \quad xC(x)^2 - C(x) + 1 = 0.$$

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**References.**

[1] Mark Haiman, *Notes on ordinary generating functions*, Math 172, Spring 2010, Sections 2–3 (PDF pages 3–5). Original notes.

[2] N. Chomsky and M. P. Schützenberger, *The Algebraic Theory of Context-Free Languages*, Studies in Logic and the Foundations of Mathematics **35** (1963), 118–161. doi:10.1016/S0049-237X(08)72023-8.

[3] Philippe Flajolet and Robert Sedgewick, *Analytic Combinatorics—Symbolic Combinatorics*, May 2002, Proposition I.6, printed page 54. Author-hosted text.